



Introduction to Quantum Computing

Embark on the quantum journey, from foundational principles to future possibilities

Quantum computing's journey starts in the early 20th century with the development of quantum mechanics. Scientists like Max Planck, Albert Einstein, and Niels Bohr began exploring the peculiar behavior of energy and matter at atomic and subatomic levels, laying the groundwork for quantum physics. The field's theoretical underpinnings were further developed through the contributions of Erwin Schrödinger, Werner Heisenberg, and Paul Dirac, among others.

The concept of quantum computing emerged in the late 20th century, notably with Richard Feynman and David Deutsch.

Feynman proposed that quantum systems could be simulated efficiently using computers operating on quantum principles, while Deutsch described the quantum Turing machine, establishing the theoretical basis for quantum computing.

The narrative then shifts to the technological advancements and experimental breakthroughs of the 21st century. This includes the development of the first quantum bits (qubits), the implementation of quantum algorithms, and the construction of early quantum processors by organizations like IBM, Google, and university research labs. Highlight significant milestones, such as Google's claim of achieving quantum supremacy in